
Feature Prediction and Gait Laterality: A Geometry Audit of Skeleton JEPA

Anonymous Author(s)

Affiliation

Address

email

Abstract

Human movement provides a useful test of whether predictive representations preserve the geometry of an articulated body. Left-right reflection supplies a known transformation, while differences between the two sides provide an observable quantity whose sign should change under that transformation. We study this relationship in a skeleton Joint-Embedding Predictive Architecture using pose sequences from gait videos. A source-separated evaluation compares trained and initial encoders, tests reflection behavior, and changes the anatomical and temporal placement of hidden training targets. The available experiments show that predicting a clip’s hidden features more accurately than another clip’s features can coexist with weaker recovery of a signed movement contrast. Motion-sensitive summaries improve the initial-encoder readout, while motion-weighted and connected-region masks provide no demonstrated advantage over their matched random references. We distinguish this evidence from exact symmetry imposed at the output and from synthetic demonstrations of explicit reflection training. The study contributes an evaluation of articulated-body representations for Physical World AI, identifying a gap between feature prediction and the recovery of a coordinate-derived movement contrast.

1 Introduction

Walking provides a useful setting for studying predictive representations because an articulated body combines anatomical regularity with movement that can differ between its sides. Exchanging left and right preserves some quantities, such as total movement, while reversing a signed left-minus-right contrast. A useful representation should retain the information needed for the intended measurement while responding predictably to changes of coordinates.

This distinction has a concrete health motivation. Post-stroke gait can exhibit spatial or temporal asymmetry, Parkinson’s research examines unequal arm swing, and cerebral palsy includes unilateral and bilateral presentations. Muscular disorders can also change pelvic and limb coordination. These clinical observations motivate examining bilateral structure without making a signed speed contrast a general measure of health. Post-stroke gait [1], arm-swing study [2], cerebral-palsy gait study [3], muscular-dystrophy kinematics [4]

We study a skeleton Joint-Embedding Predictive Architecture, or JEPA: an encoder and predictor learn to estimate hidden features using the remaining observations, while a slowly updated encoder supplies their targets. Our initial null is that this pretraining provides no gain over matched initial weights in recovering a signed movement contrast on excluded source videos. Further comparisons ask whether changing the anatomy, motion content or temporal arrangement of hidden targets produces such a gain.

The resulting investigation connects a known coordinate transformation, a target with a defined sign response, and a paired training comparison. We first examine the measurement and reflection rules, then test whether anatomically motivated training changes improve its recovery, and finally compare the predictor’s feature correspondence with the frozen encoder’s usefulness. The sequence is a retrospective account of development; it does not imply that all later questions were preregistered together.

The workshop contribution concerns evaluation of articulated-body representations. Its main evidence is that successful correct-clip feature prediction coexists with poorer laterality readouts after pretraining. The study has no demonstrated real-data forecasting or multimodal-fusion result. Those boundaries matter when considering this representation as a component of a physical world model.

2 Context in predictive and geometric learning

S-JEPA [5] establishes skeleton latent prediction, and MAMP [6] motivates motion-informed masking. The recent SLiM preprint [7] includes connected anatomical masks. Our MAMP-style arm adapts its official-code sampler while retaining the JEPA feature target; it does not reproduce the full MAMP method.

seq-JEPA [8] studies invariant and equivariant representations, while Soft Equivariance Regularization [9] examines geometric supervision and its tradeoffs. These precedents motivate evaluating geometric agreement together with observable utility.

Recent V-JEPA 2.1 [10] and Human-JEPA [11] preprints investigate dense supervision and human-focused anticipation, respectively; GaitJEPA [12] already applies predictive learning to gait silhouettes. The novelty here is the controlled evaluation of a side-sensitive coordinate target, rather than the combination of gait and JEPA or the invention of anatomical masking.

3 Measurement, training and evaluation

3.1 Cohort and target

The local GAVD [13] inventory contains 666 annotations and 642 matching pose archives. Quality control retains 625 clips from 93 source videos. Its categories are normal, Parkinson’s, stroke, myopathic and cerebral-palsy gait. They stratify source assignments and describe the dataset; they supply no encoder supervision. The category census and clinical interpretation limits appear in Appendix E.

Coordinates use image-normalized horizontal and vertical positions and inferred depth scaled relative to crop width. Their Euclidean differences are coordinate-derived movement measures, not calibrated metric three-dimensional speeds. Aspect ratio, projection, detector error and visibility can affect them.

For each of five bilateral pairs—shoulders, knees, ankles, heels, and foot tips—we calculate left and right speeds using observed timestamps and only transitions for which both landmarks are observed at both ends. Let their median speeds be $m_{L,k}$ and $m_{R,k}$. The target is

$$y(x) = \frac{1}{5} \sum_{k=1}^5 \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + \epsilon}.$$

All five pairs must provide at least eight shared valid transitions. Hips establish the pelvis reference but are excluded from this target: pelvis centering makes their speed magnitudes identical. The target is calculated before input interpolation and resizing. It measures asymmetry of median coordinate speed, not stride duration, phase coordination, or limb loading. Oppositely signed pair contrasts can cancel in the average. Similarly, left- and right-affected people could have substantial individual asymmetry but a near-zero group-average signed target. A small y therefore does not establish healthy or bilaterally symmetric gait.

Reflection M negates the centered horizontal coordinate and exchanges all anatomically paired landmarks, together with validity. Consequently, $M^2x = x$ and $y(Mx) = -y(x)$. This identity is a

property of the coordinate operation and formula, rather than evidence that every reflected recording would occur naturally.

3.2 Preservation and information boundaries

The encoder input is prepared separately. Visibility at least 0.45 and finite coordinates determine observations; up to four missing samples can be interpolated for input only. Pelvis normalization removes translation and applies a body scale. Resizing then samples 64 locations along normalized sequence index, without preserving the original timestamp intervals. Values under invalid entries are zero-filled only alongside an explicit validity mask.

Stage	Array shape	What remains associated; what changes
Archived clip	$T_i \times 33 \times 4$, plus frame numbers and frame rate	Coordinate channels and visibility remain linked to one source and sequence.
Target branch	$T_i \times 33 \times 3$, validity $T_i \times 33$	Original timestamps and observed-only paired transitions determine the target.
Model input	$64 \times 33 \times 3$, validity 64×33	Joint identities remain; interpolation, normalization and resampling alter coordinate values and timing.
Four-frame patches	$16 \times 33 \times 12$	Each patch concatenates four three-coordinate observations of one landmark; complete validity determines eligibility.
Embedded token grid	$16 \times 33 \times 96$	Time-block and landmark indices identify each 96-dimensional feature.
Batch with mask	$B \times 528 \times 96$, plus target indices and validity	The full grid remains allocated. Target selection and validity determine loss contributions; paired arms match counts per clip.

The pipeline preserves the correspondence between recordings, landmarks, masks, and targets. It cannot promise preservation of the exact observed shape or speed after preparation. Cohort preparation occurs before split-file creation in the implementation; it uses deterministic clip-local operations rather than statistics fitted across sources. Source membership is fixed before training sampling, augmentation, and model fitting.

3.3 Where laterality enters training

Five outer folds hold out 19, 19, 19, 18 and 18 sources, containing 189, 182, 72, 77 and 105 clips. All clips, mirrors and augmentations from a video remain together. Each fold and seed starts a new encoder using only its outer-training sources; source-balanced sampling controls the contribution of videos with many clips. Four inner folds in the original audit and three in the extensions select ridge readout penalties. Each imputer and scaler is fitted within the corresponding inner fitting partition, then refitted with the selected penalty on all outer-training sources. Inner validation selects the readout only: its unlabeled clips already entered that outer fold's encoder pretraining.

All completed real-data comparisons use 64 frames and 33 landmarks, width 96, encoder/predictor depths 4/2, four attention heads, batch size 20, and 1,200 updates. The online encoder retains the full token grid but zeros hidden coordinate embeddings before adding positions. The predictor estimates teacher distributions at hidden locations using cross-entropy. The teacher receives no gradients and follows an exponential moving average of online weights. An auxiliary VICReg loss, weighted by 0.05, regularizes agreement and feature variation over two unmasked geometric views pooled at twelve gait landmarks. Thus changing hidden-target eligibility does not remove all anatomical priors.

Paired arms share initialization, source draws, geometric views and update counts, with separate random streams for masking. Mask losses preserve each clip's own target indices; the latest variable-count implementation averages within clip before averaging clips. Invalid targets do not contribute. The regularizer legitimately receives complete training views, so the masked prediction pathway is not the encoder's only exposure to those observations.

The original reflection variant mirrors training clips with probability 0.5. The latest motion/region grid uses neither reflection augmentation nor an explicit reflection penalty. Notebook 09 implements the latter through two original/reflected online-encoder passes, aligning anatomical tokens while

Where laterality enters a skeleton JEPA

Encoder updates use training sources; side identity enters through joints and masks.

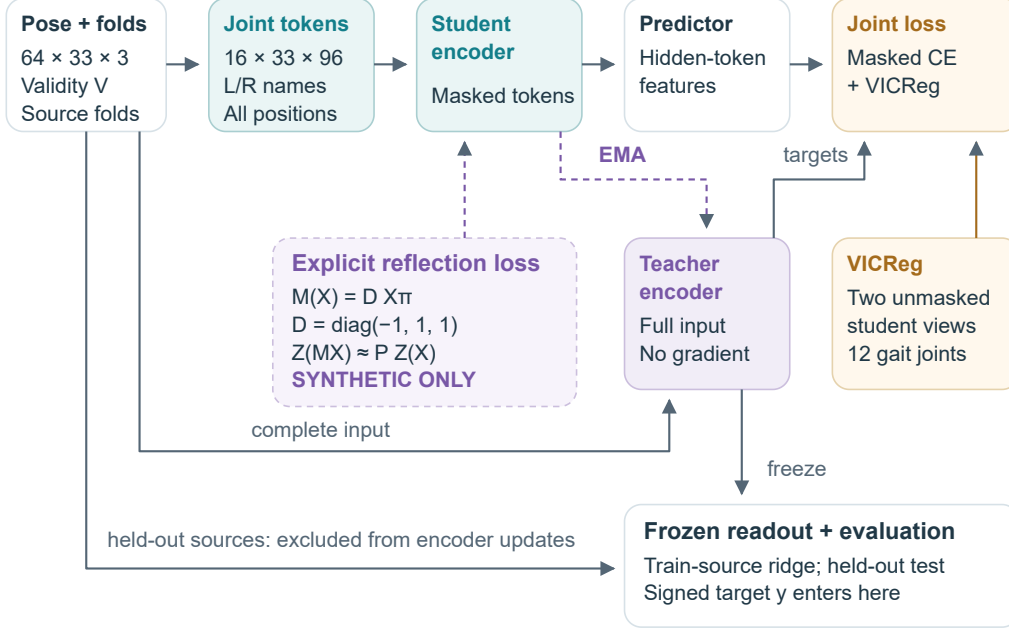


Figure 1: Anatomical identities and gait-based masking or pooling introduce laterality into the completed representation-learning recipe. The dotted branch adds an explicit token-reflection penalty to the online encoder and has synthetic evidence only. The signed target is used after pretraining by the frozen-feature readout. Clip-local preparation precedes source-fold creation in the code; every subsequent training draw and transformed view inherits the source assignment. The expanded pipeline (Figure 4) details the shapes and gradient paths.

118 leaving feature channels unchanged. Its executed example is synthetic; Appendix F gives its loss
 119 and additional compute. None of these encoder objectives uses the signed target or an affected-side
 120 label.

121 3.4 Evaluation and statistical inference

122 For V sources and n_v clips from source v , each clip has weight $w_{vi} = 1/(Vn_v)$. With $\bar{y}_w =$
 123 $\sum w_{vi} y_{vi}$,

$$R_w^2 = 1 - \frac{\sum_{vi} w_{vi} (y_{vi} - \hat{y}_{vi})^2}{\sum_{vi} w_{vi} (y_{vi} - \bar{y}_w)^2}.$$

124 Each source therefore receives equal total weight even when it supplies many clips. Within a seed,
 125 predictions from all five held-out folds are pooled before computing R^2 ; the reported result averages
 126 the five seed-specific scores. It is neither an average of fold R^2 values nor a score from an ensemble
 127 of seed-averaged predictions. Negative R^2 means squared error exceeds variation around the
 128 weighted evaluation mean; a separately fitted training-source mean is the usable prediction baseline.

129 Paired uncertainty resamples entire videos while retaining their clips and paired seed predictions.
 130 For a mask comparison the null is no positive difference in source-balanced readout performance rel-
 131 ative to the count-matched random policy; for learning it is no positive difference relative to matched
 132 initialization. These are superiority questions, and no equivalence margin or power guarantee is
 133 established. The latest mask contrasts and this revision’s learning contrasts use 2,000 resamples.
 134 Their intervals condition on the fitted models and exclude full-retraining and new-split uncertainty.
 135 The historical original summary records a separate 2,000-resample analysis, whose complete inputs
 136 remain unavailable. Repeated seeds and diagnostic mask draws are not independent participants.

Subsequent experiments were informed by the same development cohort, so their source separation does not make the overall research trajectory an untouched confirmatory study.

4 Results as a sequence of tests

4.1 Measurement agreement and the original reflection question

Recomputing the target on prepared input provides finite paired measurements for 623 clips from 92 sources. The source-weighted sign agreement with the original measurement is 70.4%, and direct calculation-agreement R^2 is 0.218. This newly repeated descriptive diagnostic flags a mismatch in the measurement path. It neither localizes the responsible preparation step nor estimates a model’s accuracy or information ceiling.

The historical reflection audit reports no demonstrated trained-over-initial predictive gain. Its documentation also records a small improvement in the specified token discrepancy with reflection augmentation: augmented-minus-vanilla $q = -0.00843$, interval $[-0.01020, -0.00687]$. Prediction does not show a corresponding demonstrated improvement. Both trained variants remain worse than their initialization under that token action. This geometric success is therefore limited to one measured criterion.

The original reports and predictions are absent from this checkout, while rounded figure values and a more precise historical README ledger remain. We use them as historical context, with the exact available values and their provenance in Appendix G. Their full uncertainty analysis was not repeated here. The direct test fixes the channel action to identity; failure does not rule out another learned reflection transformation. Likewise, an algebraically odd output or constant nonzero token vector can satisfy a symmetry criterion without useful movement prediction.

4.2 Do different hidden targets improve the observable outcome?

Notebook 12’s retained summary compares gait-only and all-landmark scattered targets at matched counts and finds no demonstrated advantage. Its raw prediction directory is also absent locally, so its numerical results remain supporting context. The main evidence comes from the complete Notebook 18 grid: 50 paired fold/seed jobs, containing 125 encoders across motion and region comparisons.

The latest motion budget uses 0.5 of the smaller gait-derived pool, yielding an average of 80.771 targets, about 17% of valid all-landmark tokens. It does not hide half the body grid. MAMP-style and robust motion weighting increase target-motion enrichment under the common audit statistic. Connected regions hide about 9.9% of valid tokens, reducing immediate visible time brackets from 0.699 to zero and visible anatomical neighbors from 0.988 to 0.512. These are successful changes to selection and local context; they do not establish what the contextualized teacher features represent.

Alternative minus its own random reference	ΔR^2	95% source interval
MAMP-style motion weighting	-0.0008	$[-0.0204, 0.0153]$
Robust motion mixture	0.0005	$[-0.0155, 0.0152]$
Connected region	-0.0087	$[-0.0333, 0.0177]$

Each comparison covers 625 clips, 93 sources and five seeds. Motion and region families have different budgets and separate references. Their intervals allow small improvements or losses, establishing neither superiority nor equivalence. Whole trajectories and interior gaps have coverage audits but no completed training comparison. No mask is selected as best on these outer-test results.

4.3 Does pretraining help when the readout retains more movement statistics?

A mean summary combines left-minus-right and left-plus-right feature means for five pairs, giving 960 features. Adding temporal standard deviations, mean absolute feature increments and ten support fractions gives 2,890. The latter improves the initial encoder from $R^2 = 0.0708$ to 0.2225, while trained teachers reach only 0.1008–0.1142 under the same summary. Every trained arm has lower R^2 and higher mean absolute error than its matched initial encoder in all five seeds. Final online encoders are also below initialization.

180 The term “initial” removes only S-JEPA pretraining. The control retains the learned pose detector,
181 anatomical schema, preparation, engineered summary and supervised ridge fit. Its improved readout
182 demonstrates access under that procedure. Temporal SD does not encode ordering, and support
183 features can describe missingness; the change in feature count also affects supervised capacity and
184 regularization.

185 We recomputed paired trained-minus-initial intervals from saved predictions during this revision,
186 without retraining. These exploratory, marginal intervals lie below zero for all five arms. For the
187 motion-uniform reference, the R^2 difference is -0.1088 , with interval $[-0.1701, -0.0412]$. Ap-
188 pendix A gives all contrasts, their bootstrap seed and their conditioning. This is evidence of a deficit
189 for the tested learning/readout combination, rather than proof that all motion information was de-
190 stroyed.

191 Ridge regularization remains unresolved: 39% of teacher and 77% of online fits using the enhanced
192 summary select the largest candidate, 10,000. A wider common search and summary-component
193 controls could change the estimates. The direct-pose summary gives $R^2 = 0.0345$; it is one linear-
194 readout baseline, not the best possible use of the coordinates.

195 4.4 What did the predictor learn?

196 Every trained predictor diagnostic favors its correct clip’s hidden teacher features over a valid mis-
197 matched target from another source, using the same control clips and positions. Initial controls
198 do not show a consistent preference. Since the teacher features are contextualized using the full
199 clip, this correspondence can depend on information already visible in the context. Static posture,
200 viewpoint, support or motion may contribute; the diagnostic does not isolate recovery of withheld
201 movement.

202 Each arm predicts its own teacher’s feature distributions. A lower loss can accompany changed target
203 variation, so neither raw cross-entropy nor between-arm feature-error magnitudes rank physical use-
204 fulness. The shared coordinate-derived endpoint provides the complementary evaluation. Recorded
205 diagnostics do not establish complete feature collapse, and the present results do not identify the
206 mechanism of the learned readout deficit.

207 5 Discussion and a decisive next experiment

208 The investigation narrows the question at each stage. The known reflection operation supplies a pre-
209 cise measurement relation; prepared-input disagreement limits attribution of an early readout failure.
210 Count-matched masking changes the selected motion and local context, yet produces no demon-
211 strated laterality benefit. A richer readout improves initialization more than trained representations,
212 even while predictors acquire correct-clip correspondence. These findings motivate investigating
213 which parts of the observable movement remain accessible through the entire measurement, repre-
214 sentation and readout procedure.

215 The most economical follow-up uses saved encoders to isolate support fractions, match summary
216 dimensions, expand ridge penalties, and compare target calculations after each preparation step.
217 Appendix B sets out competing explanations and outcomes that would weaken them. A pilot used
218 to choose a pretraining recipe must exclude its validation videos from that encoder’s training, not
219 only from its readout fit.

220 After fixing that evaluation, compare the same recipe with and without explicit reflection training.
221 Success should require improved observable prediction over both the matched base recipe and initial-
222 ization, the intended transformation behavior, and retained feature variation. A lower q alone would
223 leave the central utility question open. Measure the extra encoder passes and keep failed outcomes.
224 The current development results can guide that design; a new source collection or independently
225 specified setting is needed for stronger confirmation.

226 Future prediction would provide a more direct link to temporal world models. Prepare the input
227 from an observed prefix only, predict future teacher features, and decode common future coordinates
228 using a training-fitted decoder. Compare persistence, recent velocity, direct past coordinates, initial
229 encoders and mismatched-future training at identical eligible endpoints. Observed-future features
230 test decoder expressiveness with access to the answer period; they are not a deployable baseline.

Learned correspondence and laterality readout diverge

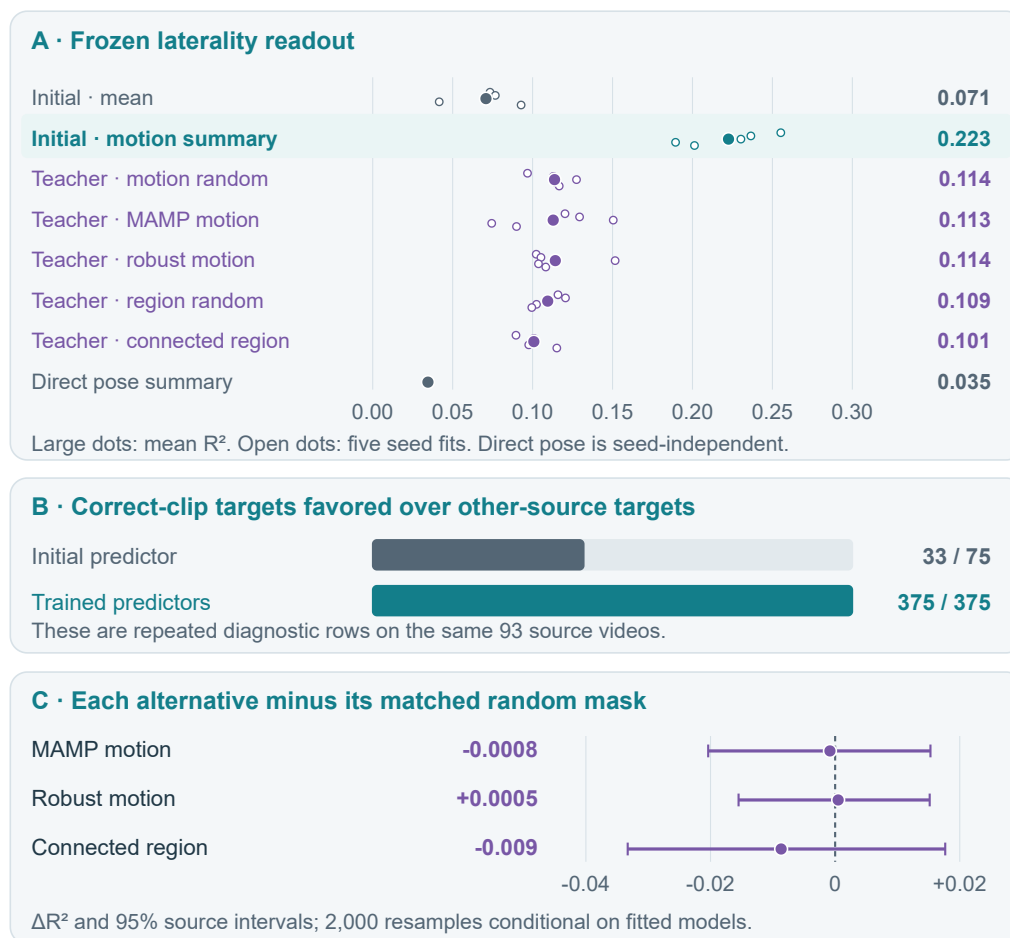


Figure 2: Predictors improve clip correspondence while their frozen encoders produce weaker laterality readouts than matched initialization. Seed points are repeated fits on the same 93 videos, not independent cohorts or confidence intervals. The 375 trained diagnostic rows are 125 fits evaluated under three masks; reused initial controls are deduplicated. The Panel C intervals are the retained mask-versus-reference source bootstraps, distinct from Appendix A's new learning contrasts. All plotted values come from the latest saved grid.

231 Appendix C illustrates the current synthetic forecast failure. No real-data forecast or recursive rollout
232 is established here.

233 **6 Scope, ethics and reproducibility**

234 The supported scope is post-development, within-GAVD performance on excluded source videos.
235 Repeated people across videos cannot be ruled out. The target has no independent clinical validation,
236 omits temporal phase and loading, and may cancel different pairwise asymmetries. Video-derived
237 coordinates and visibility are one observation pipeline; the study does not demonstrate multimodal
238 sensing, calibrated 3D reconstruction or physiological simulation.

239 The present review independently reproduces the latest grid’s 125,000 prediction rows, 200 pooled
240 score rows and three saved mask intervals, and verifies grid-table hashes and 125 complete update
241 histories. It does not claim to rehash every model checkpoint or reproduce absent historical reports.
242 The core audit, extension audit, recomputation script and figure provenance distinguish those evi-
243 dence levels. Existing research files remain unchanged.

244 Institutional ethics, data-use and derived-pose release reviews remain unresolved in the governance
245 record. No authorization for submission or artifact release is inferred from public video availability.
246 These revisions are internal manuscript artifacts.

References

- [1] K. K. Patterson, I. Parafianowicz, C. J. Danells, V. Closson, M. C. Verrier, W. R. Staines, S. E. Black, and W. E. McIlroy. Gait asymmetry in community-ambulating stroke survivors. *Archives of Physical Medicine and Rehabilitation*, 89(2):304–310, 2008. doi:10.1016/j.apmr.2007.08.142.
- [2] M. D. Lewek, R. Poole, J. Johnson, O. Halawa, and X. Huang. Arm swing magnitude and asymmetry during gait in the early stages of Parkinson’s disease. *Gait & Posture*, 31(2):256–260, 2010. doi:10.1016/j.gaitpost.2009.10.013.
- [3] S. M. Brændvik, T. Goihl, R. S. Braaten, and B. Vereijken. The Effect of Increased Gait Speed on Asymmetry and Variability in Children With Cerebral Palsy. *Frontiers in Neurology*, 10:1399, 2020. doi:10.3389/fneur.2019.01399.
- [4] I. Vandekerckhove, M. Van den Hauwe, N. De Beukelaer, E. Stoop, M. Goudriaan, M. Delporte, G. Molenberghs, A. Van Campenhout, L. De Waele, N. Goemans, F. De Groote, and K. Desloovere. Longitudinal Alterations in Gait Features in Growing Children With Duchenne Muscular Dystrophy. *Frontiers in Human Neuroscience*, 16:861136, 2022. doi:10.3389/fnhum.2022.861136.
- [5] M. Abdelfattah and A. Alahi. S-JEPA: A Joint Embedding Predictive Architecture for Skeletal Action Recognition. In *Computer Vision – ECCV 2024*, volume 15090 of *Lecture Notes in Computer Science*, pages 367–384. Springer, 2025. doi:10.1007/978-3-031-73411-3_21.
- [6] Y. Mao, J. Deng, W. Zhou, Y. Fang, W. Ouyang, and H. Li. Masked Motion Predictors are Strong 3D Action Representation Learners. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10181–10191, 2023.
- [7] J. Do, Y. Chen, G. Youk, and M. Kim. Less is More: Compact-Token Masked Feature Prediction for Skeleton Representation Learning. arXiv preprint arXiv:2603.10648v3, 2026.
- [8] H. Ghaemi, E. B. Muller, and S. Bakhtiari. seq-JEPA: Autoregressive Predictive Learning of Invariant-Equivariant World Models. In *Advances in Neural Information Processing Systems*, volume 38, pages 32943–32973, 2025. doi:10.52202/085713-1104.
- [9] J. Lee, C. Kim, H. Kim, K. Lee, and J. Lee. Soft Equivariance Regularization for Invariant Self-Supervised Learning. In *Proceedings of the International Conference on Learning Representations (ICLR)*, pages 35502–35521, 2026.
- [10] L. Mur-Labadia, M. Muckley, A. Bar, M. Assran, K. Sinha, M. Rabbat, Y. LeCun, N. Ballas, and A. Bardes. V-JEPA 2.1: Unlocking Dense Features in Video Self-Supervised Learning. arXiv preprint arXiv:2603.14482v3, 2026.
- [11] H. Wei, L. Sun, and G. Zhao. Human-JEPA: A Human-Centric Vision Model that Perceives and Anticipates. arXiv preprint arXiv:2608.21160v1, 2026.
- [12] M. J. Marin-Jimenez, I. Jimenez-Velasco, and R. Muñoz-Salinas. GaitJEPA: How Far Can We Go with JEPA on Binary Silhouettes for Gait Recognition? Accepted for the *IEEE International Joint Conference on Biometrics (IJCB)*, 2026; author repository, accessed 10 September 2026.
- [13] R. Ranjan, D. Ahmedt-Aristizabal, M. A. Armin, and J. Kim. Computer Vision for Clinical Gait Analysis: A Gait Abnormality Video Dataset. *IEEE Access*, 13:45321–45339, 2025. doi:10.1109/ACCESS.2025.3545787.
- [14] Q. Xiong, Y. Liu, J. Mo, Y. Chen, L. Zhang, Z. Xia, C. Yi, S. Jiang, and N. Xiao. Gait asymmetry in children with Duchenne muscular dystrophy: evaluated through kinematic synergies and muscle synergies of lower limbs. *BioMedical Engineering OnLine*, 22:75, 2023. doi:10.1186/s12938-023-01134-7.
- [15] Google. (2026). Pose landmark detection guide. MediaPipe documentation; 33-landmark schema.

A Additional exploratory learning contrasts

These contrasts were calculated during the 9 September revision from the complete latest prediction grid. They were not part of the original registered reflection audit. Each compares the final teacher’s motion-sensitive summary with its matched initial encoder using the same readout family, clips and seed. Intervals are marginal percentile source-bootstrap intervals conditional on the fitted models; there is no family-wise multiplicity adjustment.

Training arm	Learned minus initial R^2	95% source interval
Motion uniform	−0.1088	[−0.1701, −0.0412]
MAMP-style motion	−0.1097	[−0.1651, −0.0514]
Robust motion	−0.1083	[−0.1706, −0.0423]
Region uniform	−0.1131	[−0.1662, −0.0596]
Connected region	−0.1218	[−0.1828, −0.0553]

The recomputation script checks the prediction coverage and reproduces the recorded pooled scores before calculating these contrasts. The extension evidence audit records the retained artifact paths and full-precision values.

B Competing explanations and tests that could distinguish them

Explanation compatible with the findings	Useful next test	Outcome that would weaken it
Preparation changes the relevant measurement	Recompute the target after each preparation stage on common valid transitions, with timestamps retained	Little discrepancy at every stage despite poor learned readouts
Summary choice obscures useful features	Compare mean, support-only, temporal SD, increments and their combinations on frozen initial and trained encoders	Deficit persists under adequately selected readouts with matched capacity
Ridge regularization is insufficient	Expand the penalty grid identically for all arms using inner training-source validation	Interior selected optima still produce a learned deficit
The prescribed reflection action is too restrictive	Fit a shared involutive channel map on training pairs and evaluate held-out pairs, with initial and mismatched controls	No meaningful advantage for genuine reflected pairs on held-out sources
Predictive targets emphasize another clip property	Compare local motion supervision or visible-token targets at matched budgets and measure a common observable outcome	Improved internal correspondence still fails to improve the common outcome

These are proposed discriminating tests, not explanations established by the current data. Fitting a reflection map that preserves lengths would add an orthogonality assumption; the input reflection does not guarantee that learned feature channels obey it. Any new decision based on this cohort remains development work and should precede confirmation in an independently specified setting.

C Geometric and forecasting illustrations

A simple trajectory construction explains the limits of temporal averaging. Let left positions be (0, 1, 0, 1) and right positions (0, 0, 1, 1) at equal intervals. Both means are 0.5, but median speeds are respectively 1 and 0. Exchanging the trajectories reverses their contrast while retaining the means. This demonstrates a limitation of mean raw positions. An encoder could encode movement before pooling, so it does not prove a limitation of every mean-pooled representation.

Notebook 14’s retained synthetic demonstration uses eight generated sources, six for training and two for testing, one seed, and four updates per arm. A decoder fitted on training-source future teacher features is frozen and applied to either observed or predicted future features.

Anatomical reflection reverses a signed contrast

Back-view schematic; illustrative coordinates and target values.

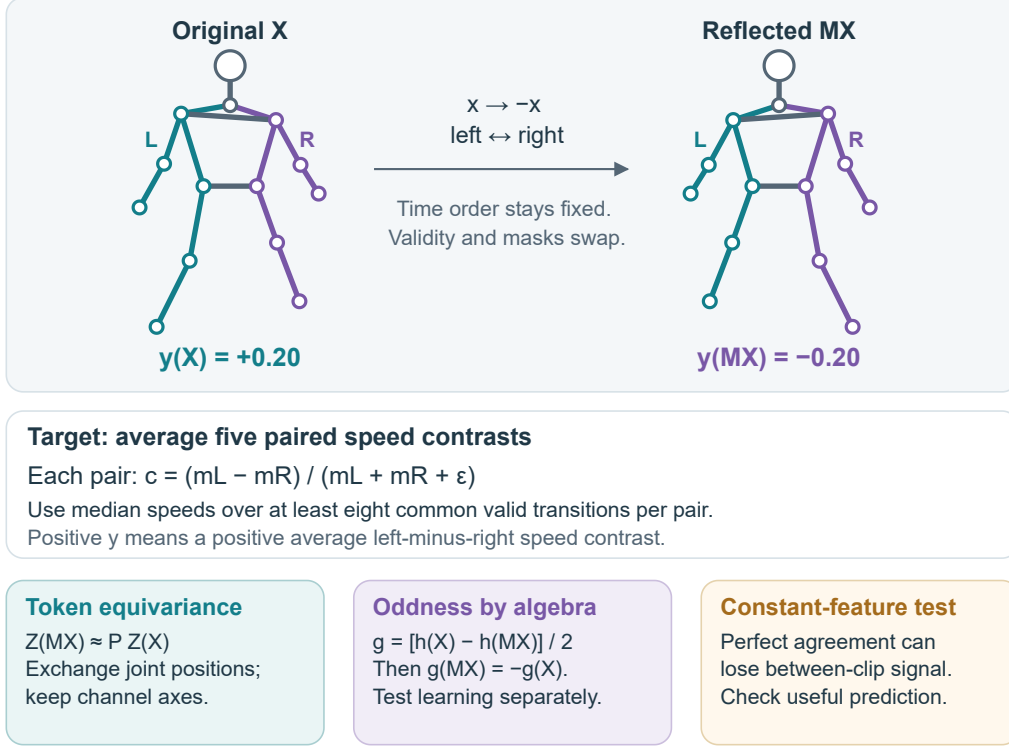


Figure 3: Reflection changes the sign of the defined target through horizontal reflection and anatomical exchange. The bodies and values are schematic, not clinical examples. An odd readout can guarantee its output transformation for any encoder. A constant nonzero token representation illustrates why geometric agreement alone is insufficient.

Future horizon	Observed-future feature RMSE	Predicted-future feature RMSE
0.25 seconds	0.011	2.51
0.50 seconds	0.013	2.61
0.75 seconds	0.025	2.64

These are displayed coordinate RMSE values in normalized coordinate units on 24 common landmark endpoints from two synthetic test clips. They are not calibrated distances. The observed-future route uses information from the answer period and tests decoder expressiveness; the predicted route is the forecast. Its failure after four updates neither establishes real gait performance nor rules out a well-trained forecasting model. Per-example artifacts for this demonstration are absent, so no new uncertainty or finer precision is inferred.

D Measurement conventions and information boundaries

Elapsed time is (frame number – first frame)/fps, and speed divides coordinate differences by positive elapsed-time increments. Target normalization uses the observed midpoint of both hips at each frame. If either hip is unavailable, that frame cannot support the target. The scale is the median of valid shoulder-pair and hip-pair widths across the clip; the implementation uses 1 when a usable scale is absent or degenerate. The target contrast uses $\epsilon = 10^{-8}$.

Input normalization separately permits a median pelvis fallback, or zero when no pelvis is observed, after short-gap interpolation. It resizes values and validity to 64 positions; validity must reach 0.999

after resizing, and invalid input values are zeroed. These are engineering conventions for noisy pose estimates, rather than a recovery of anatomical metric geometry. For past-only tasks, all such references must be estimated from the permitted prefix.

Reported prepared-coordinate corruption tests remove only small gaps after normalization and interpolation. Previously removed observations can already have influenced those operations. Those results concern representation sensitivity, not performance under genuinely absent raw observations, and are excluded from the main efficacy argument.

E Annotation census and clinical motivation

Local annotation	Clips / sources	Relevant published observation and implication
Normal	270 / 29	Supplies the dataset reference category; a zero signed contrast is not a clinical definition of normal walking.
Stroke	75 / 18	Spatial and temporal asymmetries differ across walkers; a single speed contrast covers only part of gait behavior. Patterson et al. [1]
Parkinson's	39 / 9	Arm-swing asymmetry motivates preserving side identity; our shoulder points do not measure arm swing directly. Lewek et al. [2]
Cerebral palsy	58 / 9	Studies include unilateral and bilateral presentations and examine speed-dependent symmetry. Gait-speed study [3]
Myopathic	183 / 28	Dystrophy studies examine pelvic compensation and multisegment asymmetry, illustrating features beyond this target. Longitudinal kinematics [4], kinematic-synergy study [14]

These are literature-based motivations, not findings about clinical differences in our cohort. In particular, the broad myopathic annotation does not identify Duchenne muscular dystrophy. The data do not supply verified affected-side labels, standardized disease severity, or the subtype information needed to connect target sign to a particular impairment. Here, laterality means a signed comparison of estimated left- and right-side movement.

F Training details and explicit geometric supervision

The latest recipe uses AdamW with learning rate 10^{-3} , weight decay 0.05, gradient clipping at 1, teacher EMA 0.999, student/teacher temperatures 0.10/0.06 and target-center EMA 0.9. CUDA BF16 computation retains FP32 weights, loss reductions and frozen evaluation. The original .6 gait-token mask and later .5 motion budget differ; neither is a fraction of the entire input. Valid region masks use six connected landmarks across eight time blocks, with equal realized counts in their paired reference. Equal updates do not establish equal runtime.

For clip b , define $D_b = \|Z_\theta(Mx_b) - SZ_\theta(x_b)\|_{C_b}^2$ and $E_b = \|Z_\theta(Mx_b)\|_{C_b}^2 + \|SZ_\theta(x_b)\|_{C_b}^2$, where C_b identifies common valid positions. Notebook 09 adds

$$\mathcal{L}_{\text{ref}} = \frac{1}{B} \sum_b \frac{D_b}{\max(\text{stopgrad}(E_b), 10^{-12})}.$$

This matches the implementation's detached, clamped denominator. Both numerator branches update the shared online encoder through two additional unmasked passes. The demonstration uses weight 1, 16 input frames and width 16. It has no real-GAVD result and supplies no signed target supervision. Identity action on feature channels is a modeling choice; a constant nonzero representation can satisfy it.

The expanded pipeline (Figure 4) and figure documentation explain all training and evaluation entry points. For the proposed extension, a fitted involutive feature-channel action or explicit even/odd channels would test another geometric assumption. Neither has a completed real-data result in this package.

G Historical results retained as context

The original figure JSON supplies three-decimal values, while the existing docs README retains five-decimal historical estimates and a claim of prior artifact verification. That prior verification

cannot be repeated from the current checkout because the original report/prediction/checkpoint chain is absent. No unavailable value is inferred.

Historical comparison	Estimate	Recorded 95% interval
Learned primary R^2	0.05979	[−0.02527, 0.12571]
Learned minus initial R^2	−0.01798	[−0.03851, 0.00248]
Reflection augmented minus vanilla R^2	0.00408	[−0.00556, 0.01277]
Reflection augmented minus vanilla token q	−0.00843	[−0.01020, −0.00687]
Learned minus initial token q , vanilla	0.03055	[0.01576, 0.04763]
Learned constructed-odd R^2	0.04302	[−0.04356, 0.11283]
Learned minus initial constructed-odd R^2	−0.05874	[−0.09549, −0.01740]

The token test uses $q = D/E$ after joint permutation with identity channel action. Learned vanilla $q = 0.114$, interval [0.095,0.138], is retained at three decimals; its interval crosses the operational 0.10 margin, while failing the upper-bound acceptance rule. A low q can reflect insensitive shared features.

The odd construction uses $z^-(x) = [z(x) - z(Mx)]/\sqrt{2}$, no feature centering and no readout intercept. Its sign reversal is algebraic. The reported performance tests the whole readout pipeline and cannot identify every source of its useful information.

Notebook 12’s retained teacher R^2 values are −0.022633 for gait targets and −0.010449 for all-landmark targets, with initial 0.048160 and direct pose 0.129978. The all-landmark-minus-gait contrast is 0.012184, interval [−0.018303,0.050207]. Its raw predictions were unavailable for this review. Notebook 08 used an earlier implementation and fixed penalty; its more negative numbers are omitted from the main argument. Comparing scores across these recipes cannot isolate a masking or regularization effect.

H Latest absolute results and notebook map

Frozen features or control	Mean $R^2 \pm$ seed SD	Mean absolute error
Training-source target mean	−0.0106 $\pm$ 0.0000	0.046172
Direct prepared-pose summary	0.0345 $\pm$ 0.0000	0.044960
Initial encoder, mean summary	0.0708 $\pm$ 0.0186	0.044344
Initial encoder, motion-sensitive summary	0.2225 $\pm$ 0.0268	0.041546
Motion-uniform teacher, motion-sensitive	0.1137 $\pm$ 0.0110	0.043658
MAMP-style teacher, motion-sensitive	0.1129 $\pm$ 0.0306	0.043751
Robust-motion teacher, motion-sensitive	0.1142 $\pm$ 0.0210	0.043595
Region-uniform teacher, motion-sensitive	0.1094 $\pm$ 0.0088	0.044037
Connected-region teacher, motion-sensitive	0.1008 $\pm$ 0.0092	0.043756

Each row covers 625 clips and 93 sources. SD describes the five optimization seeds, not sampling uncertainty. The initial encoder, direct-pose, and training-mean controls are reused across arms; their repeated appearance in machine-readable tables is not independent replication. “Initial” refers only to the S-JEPA weights: the pipeline still uses a pretrained pose detector, supplied anatomical identities, feature engineering, and a supervised ridge readout.

Notebooks	Question or role	Evidence used here
00–02	Protocol, target/cohort audit and source splits	Code, cohort and splits checked; current scope is internally fixed after development
03–05	Fold-local training, held-out probes and aggregation	Historical summary; original complete report chain unavailable
06	External-subject evaluation gate	No completed external validation
07	Measurement and readout diagnostics	Prepared-target agreement recomputed on 623 clips / 92 sources
08	Earlier matched-count anatomical masking	Historical context; not pooled with later recipes
09	Explicit reflection training	Synthetic only
10	Past-only movement prediction	Implemented route; no retained real training result
11–13	Mask construction, controlled training and predictor/readout checks	Notebook 12 retained real summaries; teaching executions separately synthetic
14	Decode observed versus predicted future features	Synthetic demonstration only
15–16	Motion selection and structured-context audits	Real-data intervention checks
17–18	Paired training and motion-sensitive readouts	Complete latest grid independently recomputed

Detailed anatomy-aware JEPA pipeline**A · Prepare each clip independently****Model input**

Short-gap filling; pelvis centering;
body scaling; 64-frame resizing.
X: $B \times 64 \times 33 \times 3$
Validity: $B \times 64 \times 33$

Signed movement contrast

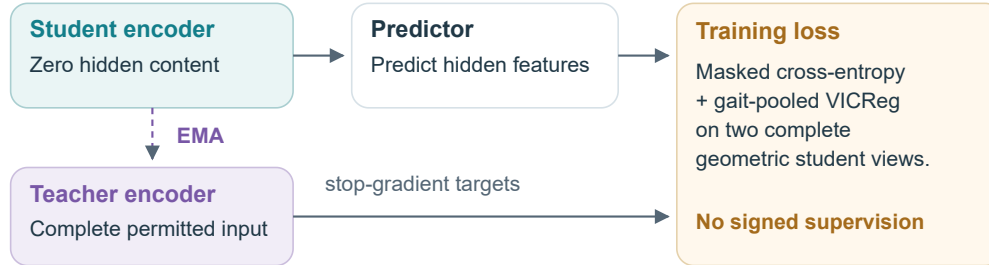
Original timestamps and valid pairs.
No filling or temporal resizing.
Five bilateral pairs define target y.
Image-derived coordinates.

B · Fix five source-video folds

625 clips / 93 sources; every clip and derived view inherits its source fold.

C · Keep the full anatomical grid; vary hidden targets

Patches: $B \times 16 \times 33 \times 12 \rightarrow$ tokens: $B \times 528 \times 96$.
12: gait vs all joints. 15–18: motion or connected-region masks.
All 33 joints stay in the input. Valid and hidden masks: $B \times 16 \times 33$.

D · Train only on outer-training sources**E · Notebook 09 adds an explicit encoder relation**

$M(X)[t, j] = (-x, y, z)[t, \pi(j)]$; compare $Z(MX)$ with $P Z(X)$.
Aligned valid tokens; identity channel action; detached energy denominator.
SYNTHETIC ONLY · 16 frames / 16 channels · two extra encoder passes

F · Freeze the encoder, then fit and evaluate the readout

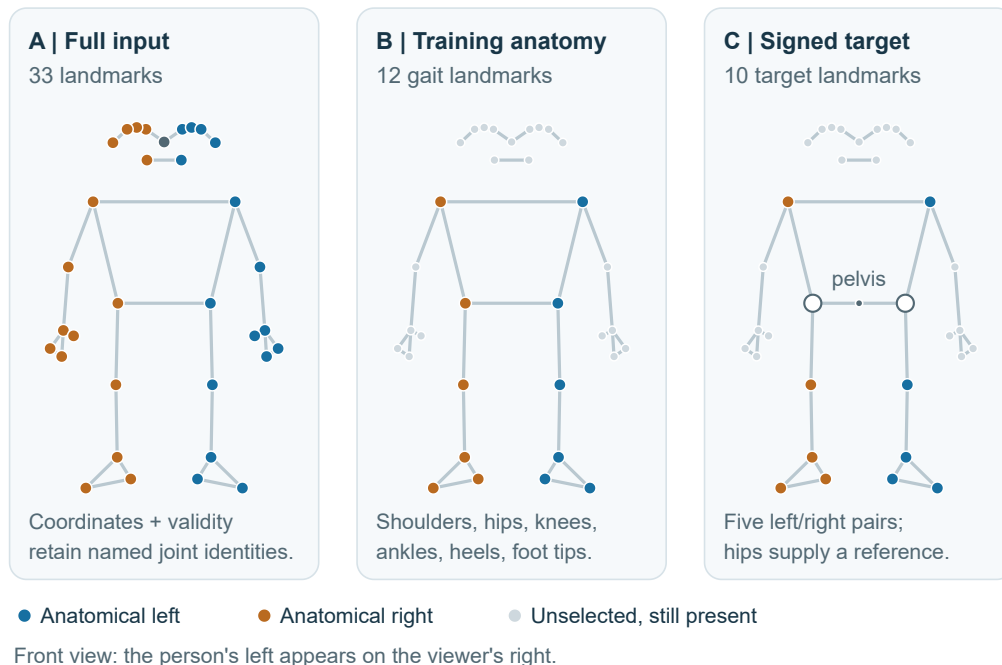
Fit scaling and ridge only on training-source groups; predict held-out videos.
Compare with initial S-JEPA weights using the same pose extractor and schema.
Pool held-out folds per seed; weight every source video equally.
Video-bootstrap intervals condition on fitted models; seeds reuse the cohort.

Preparation precedes fold-file creation; training sampling follows source grouping.
Tensor axes and joint identities persist; interpolation and resizing change coordinates.

Figure 4: Expanded version of the training pipeline linked in Figure 1, showing tensor shapes, source separation, and the distinct gradient paths for the completed training recipe and the synthetic reflection extension.

One landmark array, two anatomical selections

Highlighting selects a use for a point; all 33 positions stay in the input.



The two selections answer different questions

Training: historical target eligibility and VICReg use the 12 gait points.

Latest motion / region samplers can select other body tokens.

Target: use observed timestamps before interpolation or 64-frame resizing.

Figure 5: Landmark-selection schematic using the published MediaPipe identities [15]. All 33 positions remain allocated in the encoder input. The 12 highlighted gait landmarks identify historical target eligibility and VICReg pooling; the latest motion and region samplers can select other tokens. The signed target uses five bilateral pairs, with hips supplying the pelvis reference rather than a sixth contrast. Coordinates are hand-drawn for explanation and represent neither measured trajectories nor disease examples.